Module 5, Lesson 1

AI 101 FOR MECHANICAL ENGINEERS

Investigate before changing readings



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Lesson Structure

1. Introduction and Screening
2. Missingness and Imputation
3. Fault Event Analysis
4. Concept Reference and Assessment

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1. Introduction and Screening

Establish lesson objectives and fundamental data inspection parameters.

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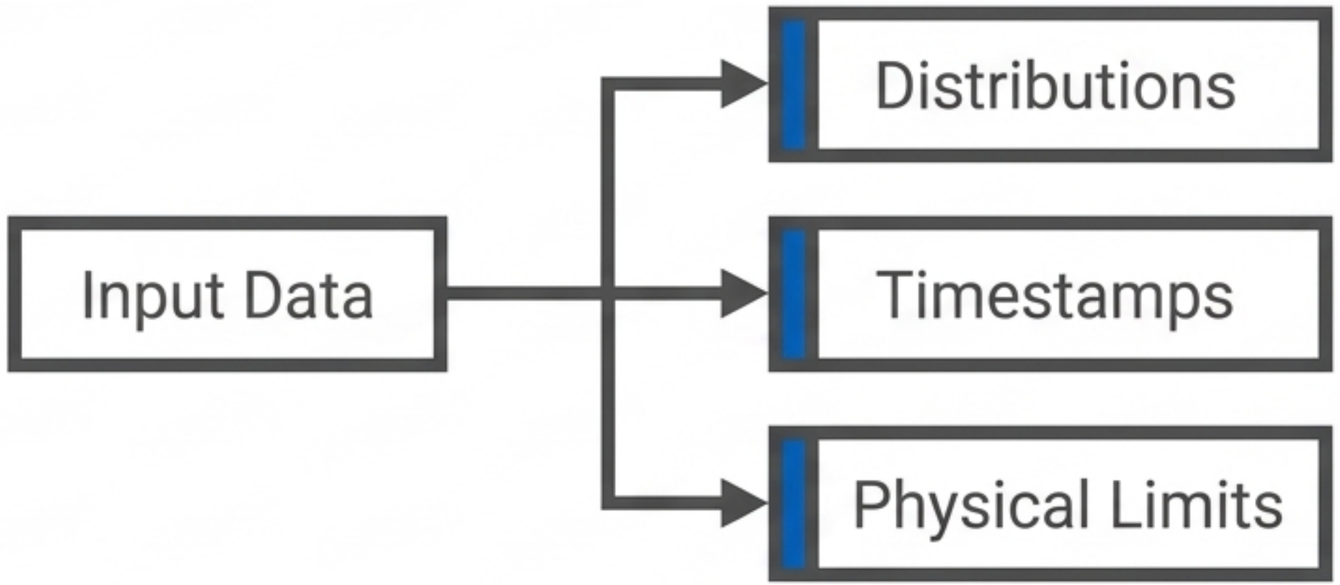
Lesson Outcomes

Investigate unusual and missing dataset measurements thoroughly.

Explain imputation effects and data recovery limitations.

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Screening Measurements



- Identify physical impossibilities like negative absolute pressure.
- Detect inconsistent input units including W versus kW.
- Retain unusual readings reflecting targeted mechanical events.
- Avoid deleting data solely to improve performance scores.

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2. Missingness and Imputation

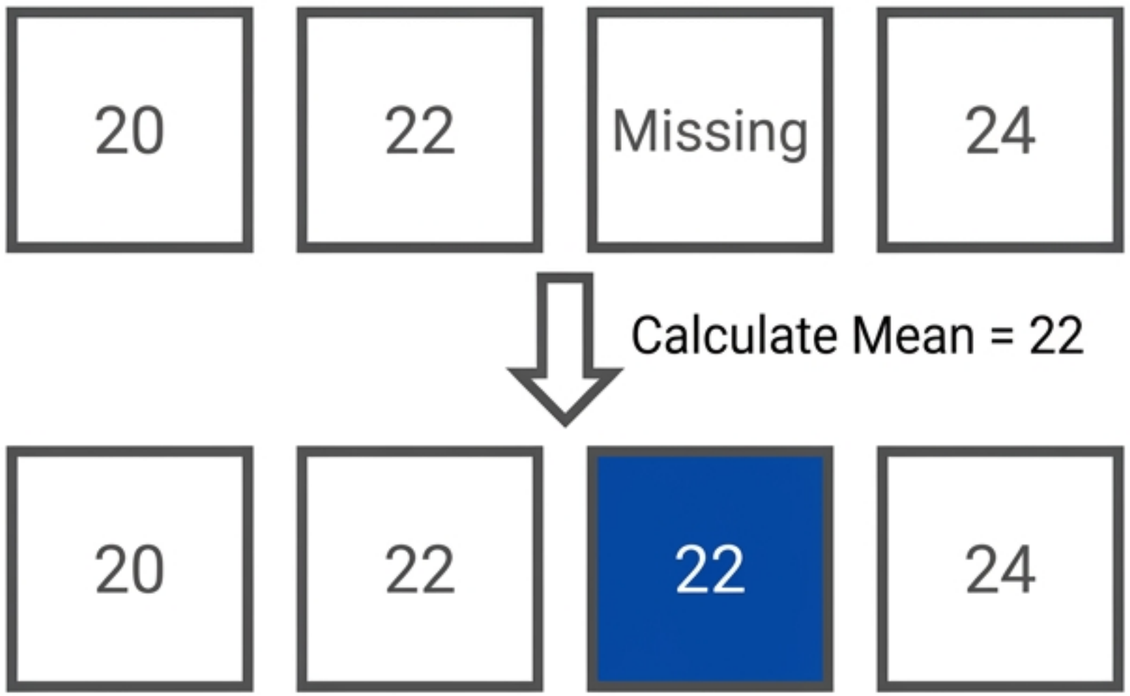
Define missingness characteristics and demonstrate the mechanics of mean imputation.

Missingness Characteristics



- Missing data indicates physical states, like sensor dropout.
- Imputation applies declared mathematical rules for replacement.
- Imputation cannot recover original physical measurement values.
- Document all modified values and applied rules explicitly.

Mean Imputation Process



- Calculate replacement values from observed dataset points.
- Replace missing features with the calculated mathematical mean.
- Imputation reduces visible dataset variability artificially.
- Estimate replacements using training data exclusively.

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Technical Vocabulary

Noise	Variation obscuring the primary quantity of interest.
Outlier	Unusual observation relative to comparison populations.
Imputation	Filling missing values via stated mathematical rules.

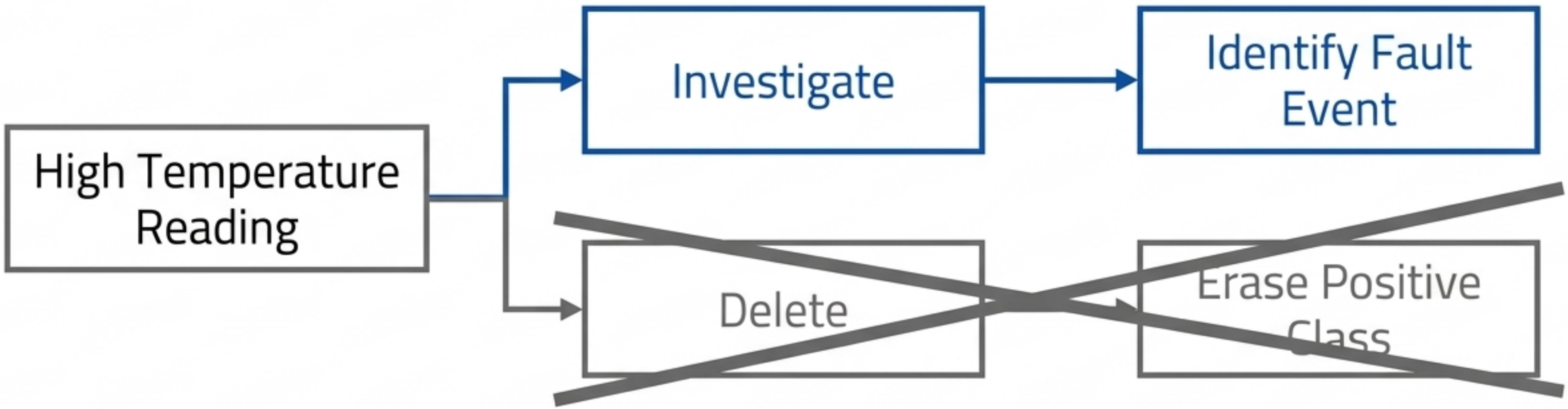
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3. Fault Event Analysis

Analyze the impact of outlier deletion on fault-detection pipelines.

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Outlier Deletion Impact



- High temperatures may represent critical mechanical fault events.
- Automated deletion removes the targeted positive class entirely.
- Premature removal generates mathematically misleading analysis datasets.
- Investigate operating context before removing any outlier data.

4. Concept Reference and Assessment

Consolidate diagnostic responses and assess technical understanding.

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Observation Response Matrix

Observation	Required First Response
Missing reading	Investigate communication loss or system dropouts.
High temperature	Retain operating context and investigate thoroughly.
Mixed W and kW	Convert to consistent SI units.
Imputed value	Mark and document changed observation points.

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Operational Protocols

Outlier Action = Investigate Context

Imputer Fitting = Training Statistics

- Investigate measurement context prior to deleting structural outliers.
- Unusual values often indicate meaningful mechanical fault events.
- Estimate learned preprocessing using training data statistics only.
- Never fit imputers using final-test input data arrays.

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Key Points

1. Screen datasets for physical impossibilities and unit inconsistencies.

2. Imputation applies rules without recovering original physical measurements.

3. Retain investigated outliers representing critical mechanical fault events.

4. Standardize analytical responses to structural system data anomalies.

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